

An Exact Schubert and Morse–Bott Atlas for the Grassmann Projection Flow

Route-A source-local certificate HCS-C298

2 September 2026

Abstract

For a real symmetric matrix A , we solve the Grassmann projection equation $\dot{P} = [P, [P, A]]$ globally by $\text{Ran } P(t) = e^{tA} \text{Ran } P(0)$. Without assuming distinct subset sums, we identify every simple-spectrum Schubert-cell limit and its actual-support exponential rate, all linear modes, and the complete repeated-spectrum product-Grassmann Morse–Bott atlas with associated-graded limits.

1 Projection dynamics and the exact quotient

Let $A = A^T \in \mathbb{R}^{n \times n}$ and $1 \leq k \leq n - 1$. We realize $\text{Gr}(k, n)$ as the rank- k orthogonal projections

$$\mathcal{P}_{k,n} = \{P : P^T = P, P^2 = P, \text{tr } P = k\}$$

and consider

$$\dot{P} = [P, [P, A]] = AP + PA - 2PAP. \quad (1)$$

The sign in (1) selects the high-eigenvalue subspaces forward in time. The generator A is fixed; the state is the subspace represented by P .

Theorem 1 (Global exact projector). *Let $Q_0 \in \mathbb{R}^{n \times k}$ have full column rank and $\text{Ran } Q_0 = \text{Ran } P_0$. For every $t \in \mathbb{R}$, set*

$$Y(t) = e^{tA} Q_0, \quad P(t) = Y(t) (Y(t)^T Y(t))^{-1} Y(t)^T. \quad (2)$$

Then (2) is the unique global solution of (1); in particular,

$$\text{Ran } P(t) = e^{tA} \text{Ran } P_0.$$

If $Ae_i = \lambda_i e_i$ in an orthonormal eigenbasis, its projective Plücker coordinates obey

$$p_I(t) = e^{t\lambda_I} p_I(0) \quad \text{up to one common scale,} \quad \lambda_I = \sum_{i \in I} \lambda_i. \quad (3)$$

Proof. The matrix e^{tA} is invertible, so $Y(t)$ has full column rank and its Gram matrix is positive definite for all real t . Formula (2) is therefore the orthogonal projector onto $e^{tA} \text{Ran } P_0$. Since $Y' = AY$ and $A = A^T$, differentiation of the Gram inverse gives

$$\dot{P} = AP + PA - 2PAP = [P, [P, A]].$$

Smooth finite-dimensional ODE uniqueness identifies it with the solution. On $\bigwedge^k \mathbb{R}^n$, the induced map sends $e_{i_1} \wedge \cdots \wedge e_{i_k}$ to $e^{t(\lambda_{i_1} + \cdots + \lambda_{i_k})}$ times itself, proving (3). $\square$

This quotient identity is the subspace power-flow mechanism neighboring Oja’s principal-component dynamics and the continuous Grassmann flows of Absil, Sepulchre, and Mahony. We derive every statement needed below and make no priority claim for that classical mechanism.

2 Simple spectrum: every Schubert cell and its rate

Assume in this section

$$\lambda_1 < \lambda_2 < \cdots < \lambda_n, \quad F_j = \text{span}(e_1, \dots, e_j). \quad (4)$$

For a k -plane V , write

$$\mathcal{B}(V) = \{I \subset \{1, \dots, n\} : |I| = k, p_I(V) \neq 0\}.$$

These are the bases of the representable matroid determined by the rows of any frame for V .

Lemma 1 (Matroid guard). *Distinct element weights $\lambda_1, \dots, \lambda_n$ give a unique maximum-weight basis $I_+(V)$ and a unique minimum-weight basis $I_-(V)$ in $\mathcal{B}(V)$, even when two arbitrary k -subset sums coincide.*

Proof. Run the matroid greedy algorithm in decreasing element weight. To see uniqueness directly, suppose two optimal bases differ and take the largest-weight element in their symmetric difference. Strong basis exchange inserts that element into the basis missing it while removing a strictly lower-weight element, producing a heavier basis. This contradiction proves the maximum statement; reverse the weights for the minimum. $\square$

For $I = (i_1 < \cdots < i_k)$, let Ω_I be the Schubert cell characterized by

$$\dim(V \cap F_j) = \#\{a : i_a \leq j\} \quad (1 \leq j \leq n). \quad (5)$$

Equivalently, I is the lexicographically greatest nonzero Plücker index, hence the greedy maximum in Lemma 1.

Theorem 2 (Exact cell limits and actual-support rates). *If $V_0 \in \Omega_I$, then*

$$\lim_{t \rightarrow +\infty} P(t) = P_I,$$

the coordinate projection onto $E_I = \text{span}\{e_i : i \in I\}$. The opposite Schubert cell indexed by $I_-(V_0)$ gives the backward limit.

If $V_0 \neq E_I$, define

$$\rho_2(V_0) = \max_{J \in \mathcal{B}(V_0), J \neq I} \lambda_J, \quad \Delta_+(V_0) = \lambda_I - \rho_2(V_0) > 0. \quad (6)$$

Then

$$\lim_{t \rightarrow +\infty} -\frac{1}{t} \log \|P(t) - P_I\|_F = \Delta_+(V_0). \quad (7)$$

The analogous backward rate uses the actual second-smallest nonzero Plücker weight. For an equilibrium the error is zero and its gap is declared infinite.

Proof. By Lemma 1, I is the unique maximum supported weight despite possible ambient subset-sum ties. Factor $e^{t\lambda_I}$ from the Plücker vector in (3). Every other supported coordinate decays, giving the coordinate-plane limit. The nonzero sum of all weight- ρ_2 coordinate vectors is the first surviving transverse term, so projective distance is $\Theta(e^{-t\Delta_+})$. The Plücker embedding and the orthogonal-projector chart are smooth local embeddings with equivalent norms; the logarithmic rate is therefore (7). Apply the argument to $-A$ for backward time. $\square$

Why the support qualifier matters. Simple eigenvalues do not imply distinct k -fold sums; for example $0 + 4 = 1 + 3$. Zero Plücker coordinates remove subsets from the orbit, and ties may remain at subleading weights. Lemma 1 proves only the needed top supported uniqueness, while (6) measures the first term that is actually present.

3 Equilibria, modes, and Schubert dimensions

Relative to $\text{Ran } P \oplus \ker P$, write the off-diagonal block of A as B . Then $[P, [P, A]]$ has off-diagonal blocks B and B^T . Thus

$$\dot{P} = 0 \iff [P, A] = 0 \iff \text{Ran } P \text{ is } A\text{-invariant}. \quad (8)$$

For simple spectrum these are exactly the $\binom{n}{k}$ coordinate projections.

At P_I , a tangent graph component from selected e_i to unselected e_j has linearized equation

$$\dot{x}_{ji} = (\lambda_j - \lambda_i)x_{ji}. \quad (9)$$

Consequently

$$d_s(I) = \#\{(i, j) : i \in I, j \notin I, \lambda_j < \lambda_i\}, \quad (10)$$

$$d_u(I) = \#\{(i, j) : i \in I, j \notin I, \lambda_j > \lambda_i\}. \quad (11)$$

If $I = (i_1 < \cdots < i_k)$, then $d_s(I) = \sum_{a=1}^k (i_a - a) = \dim \Omega_I$ and $d_s(I) + d_u(I) = k(n - k)$.

4 Repeated spectrum: associated grade and Morse–Bott strata

Let the distinct eigenvalues be $\mu_1 < \dots < \mu_s$, with eigenspaces E_α of dimensions m_α , and put $F_\alpha = \bigoplus_{\beta \leq \alpha} E_\beta$. For an initial plane V , define

$$G_\alpha = \pi_\alpha(V \cap F_\alpha) \subset E_\alpha, \quad k_\alpha = \dim(V \cap F_\alpha) - \dim(V \cap F_{\alpha-1}), \quad (12)$$

where π_α is orthogonal projection to E_α .

Theorem 3 (Associated-graded limit and complete critical atlas). *Every initial plane has the basis-independent limit*

$$\lim_{t \rightarrow +\infty} e^{tA} V = \bigoplus_{\alpha=1}^s G_\alpha. \quad (13)$$

For every feasible occupancy vector $0 \leq k_\alpha \leq m_\alpha$, $\sum_\alpha k_\alpha = k$, the corresponding equilibrium component is exactly

$$\mathcal{C}_k = \prod_{\alpha=1}^s \text{Gr}(k_\alpha, E_\alpha). \quad (14)$$

It is Morse–Bott. Its critical, stable-normal, and unstable-normal dimensions are, respectively,

$$d_0 = \sum_\alpha k_\alpha (m_\alpha - k_\alpha), \quad (15)$$

$$d_s = \sum_{\beta < \alpha} k_\alpha (m_\beta - k_\beta), \quad (16)$$

$$d_u = \sum_{\beta > \alpha} k_\alpha (m_\beta - k_\beta), \quad (17)$$

and $d_0 + d_s + d_u = k(n - k)$.

Proof. Choose a basis of V adapted to the filtration $V \cap F_\alpha$. When a basis vector first appears at level α , its E_α component is nonzero; these leading components are independent and span G_α . Under e^{tA} every lower component decays relative to that leading one, which proves (13). Equivalently, the entire highest-weight component of the decomposable Plücker vector survives. We do not choose one coordinate from a tied block. The descending eigenflag gives the backward limit.

Equation (8) and self-adjoint spectral decomposition show that every invariant plane splits as $\bigoplus V_\alpha$ with $V_\alpha \in \text{Gr}(k_\alpha, E_\alpha)$, proving completeness of (14). Within-block graph modes have rate zero and are precisely tangent to this product. Every cross-block mode has the nonzero rate $\mu_\beta - \mu_\alpha$; counting its sign gives (15). Thus the Hessian is nondegenerate normal to the critical product, which is the Morse–Bott condition. The dimension closure counts every selected-to- unselected graph coordinate exactly once. $\square$

5 Strict Lyapunov law and recurrence

Define $\Phi(P) = \text{tr}(AP)$. Cyclicity of trace and $P^2 = P$ give

$$\frac{d}{dt} \Phi(P(t)) = \text{tr}(A[P, [P, A]]) = \|[P, A]\|_F^2 \geq 0. \quad (18)$$

Equality holds exactly at the invariant projections. If a nonconstant orbit returned arbitrarily close to its initial point, continuity of Φ would force its values back toward $\Phi(P_0)$ after strict increase, contradicting monotonicity. Hence no nonconstant recurrent or periodic trajectory exists. The explicit limits above are stronger than this obstruction.

Source lineage

Oja owns the principal-component learning-flow neighborhood; Brockett owns the double-bracket sorting framework; Absil, Sepulchre, and Mahony directly treat continuous-time subspace flows related to the symmetric eigenproblem. The entries below are ownership citations, not claims of literature priority for this package.

References

- [1] E. Oja, “Simplified neuron model as a principal component analyzer,” *Journal of Mathematical Biology* 15 (1982), 267–273. DOI: [10.1007/BF00275687](https://doi.org/10.1007/BF00275687).
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- [3] P.-A. Absil, R. Sepulchre, and R. Mahony, “Continuous-time subspace flows related to the symmetric eigenproblem,” *Pacific Journal of Optimization* 4 (2008), 179–194. Institutional record: [hdl:2078.5/90452](https://hdl.handle.net/2078.5/90452).